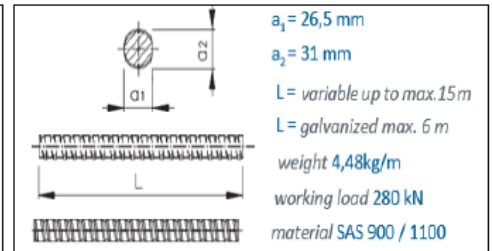
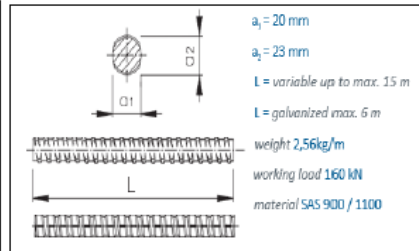
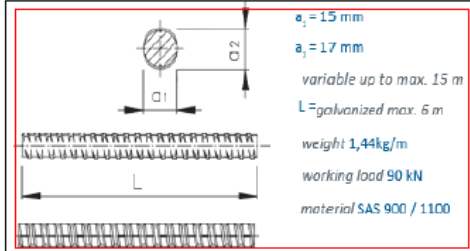


Tie rod Ø15mm

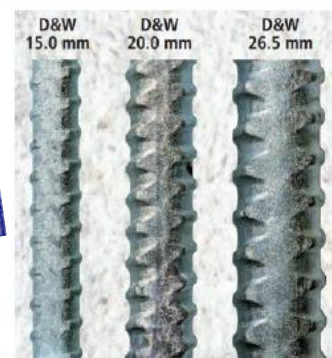
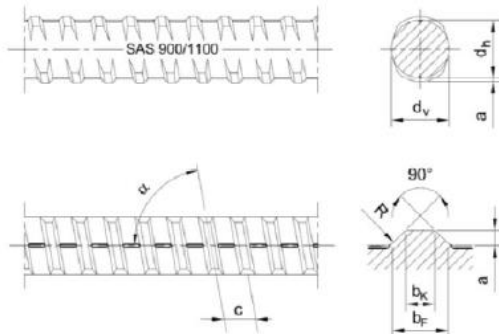
Definition of the product:

hot-rolled and heat-treated steel bar anchors St 900/1100 with thread ribs and a nearly circular cross section. The nominal diameter is 26.5 mm. Two opposing rib rows are rolled onto its surface to form a single right-hand thread. Formwork Ties offers in 15 mm (5/8"), 20 mm (7/8") and 26.5 mm (1") diameters.



yield stress / ultimate stress	nom.-Ø	working load	yield load	ultimate load	cross section area	weight	elongation		
	[mm]	[kN]	[kN]	[kN]	[mm²]	[m/to]	[kg/m]	$A_{e1} [\%]$	$A_{10} [\%]$
SAS 900/1100 FA grade 160 FA					warmgewalzt, schweißbar hot rolled, weldable				
	15	90	159	195	177	694,4	1,44	3	7
SAS 900 / 1100 Typ FA	20	160	283	345	314	390,6	2,56		
	26,5	280	495	606	551	223,2	4,48	2	7

Shaping



Nominal diameter	Nominal weight ¹⁾	Nominal cross section	Core diameter		Thread ribs (right-hand)					
					Height	Width (foot)	Width (head)	Spacing	Pitch	Radius
Øp = dp	g	Ap	dh	dv	a	b _F	b _K	c	a	R
mm	kg/m	mm²	mm	mm	mm	mm	mm	mm	degree	mm
15,0	1,41	173	14,8	14,7	1,15	4,8	2,5	10,0	78,5	1,5
20,0	2,51	309	19,8	19,6	1,30	4,8	2,2	10,0	81,5	2,0
26,5	4,41	542	26,3	26,0	1,60	6,4	2,6	13,0	81,5	2,6

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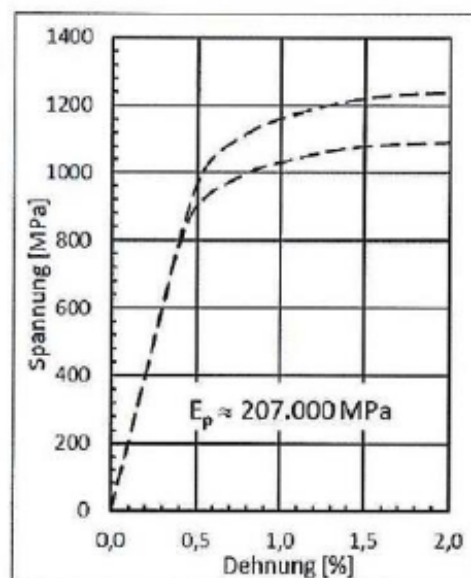
Tie rod Ø15mm

Table: Strength and deformation properties

1		Strength classes of steel bar with thread ribs		St 900/1100	Quantiles ^{a)} [%]
2	0,2 % yield point	R _{p0,2}	[N/mm ²]	900	5,0
3	tensile strength	R _m	[N/mm ²]	1100	5,0
4	Traction at 0,2% strain Ø _p 26,5 mm	F _{p0,2}	[kN]	488	5,0
5	max. tensile load Ø _p 26,5 mm	F _m	[kN]	596	5,0
6	max. drop in load capacity (T ^{b)}) after bending back and forth through 180° (Bending roll diameter: d _{br} = 6 x d _p) Acc. DIN 488-1:1984-09	T	[%]	10,0	-- ^c
7	Elongation at break	A _{11,3}	[%]	7,0	5,0
8	Total elongation at maximum force	A _{gt}	[%]	3,0	5,0
9	Mandrel diameter for bending test acc. DIN EN ISO 15630-3:2020-02, Section 6 with a bending angle of 180 °	ØD _m	[mm]	6x d _p	-- ^c
10	Impact work Acc. DIN EN ISO 148-1:2011-01 at -20°C	KV ₂	[J]	27	-- ^c

a) Quantiles for a statistical probability of W = 1 - α = 0,95 (one-sided)
b) When T = (1 - R_{m,ist after bending} / R_{m,ist}) x 100
c) Each single value

Picture 2: Principal stress-strain curve



The lines give an orientation for the basic stress-strain curve assuming the calculated value of the E-module (E_p) of 207,000 N/mm²

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Water Barrier Ø15mm

Definition of the product:

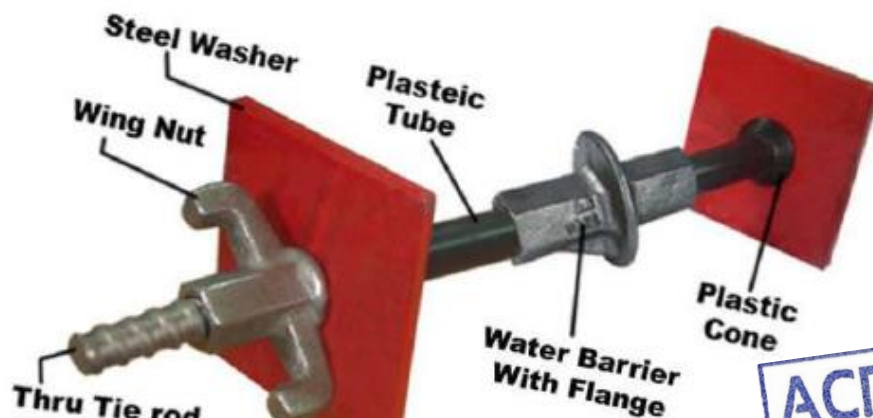
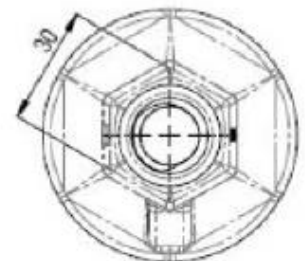
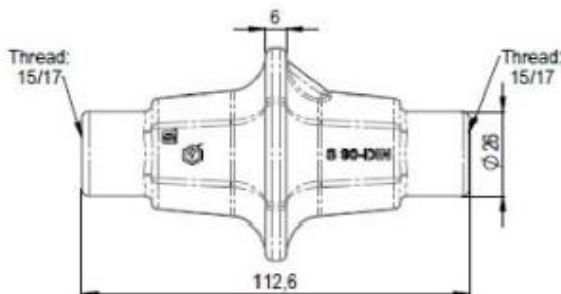


The water barrier is a sealing sleeve made from cast iron with hexagonal section, a central stop having at both sides a special internal thread for DWG tie rods.

The sleeve presents a circular flange in the middle of its length, lengthening the path for water which would eventually seep alongside the thread bars or the tubes.

Properties

Mechanical properties	
PC15MWF - Waterstop sleeve for DWG tie rods Ø15.1mm	
Service load	90 kN
Testing load	190 kN
Resistance to water pressure	7 bar (= 0.7 MPa)
PC20MWF - Waterstop sleeve for DWG tie rods Ø20.0mm	
Service load	150 kN
Testing load	345 kN
Resistance to water pressure	7 bar (= 0.7 MPa)



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